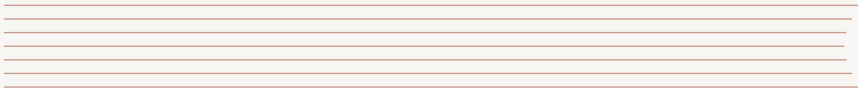


DATA WAREHOUSE

Air Quality, Economy and Health in Europe (2010–2020)

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midcentury
modern



OUTLINE

- Introduction and Dataset
- Conceptual Design (DFM)
- Starflake Logical Schema
- Reconciled Layer: 3-Tier ELT
- OLAP Analysis and Insights
- Conclusions

■ The starting idea

Is there an observable relationship between **air pollution**, **economic wealth**, and **mortality** across European countries? Answering this requires a schema able to connect different domains on the same geographic and time basis — this is why we chose a Data Warehouse.

Environment (Air Quality), Economy (GDP), and Health (Mortality) are usually analyzed separately: combining them into a single multidimensional model enables comparisons that would otherwise be impossible — for example, understanding whether a richer country also breathes cleaner air and lives longer. Ten-year coverage across Europe (2010–2020).

DOMAINS AND DATA SOURCES

Area	Source	Files	Format
Air Quality	WHO	1 (city, PM2.5/PM10/NO2)	Excel
Mortality (HFAMDB)	WHO	4 (<i>All causes</i> + 3 specific causes)	CSV
Economy (GDP)	Eurostat	1	TSV
Population	Eurostat	1	TSV

Even data from the same source come in different formats (GDP and Population are 2 separate Eurostat files, not one): structures, country codes, and granularity do not match — a cleaning and integration step is needed before loading them into the same Data Warehouse.

A CLOSER LOOK AT THE SOURCE FIELDS

WHO — Air Quality (Excel)

IS03, WHO Country Name, City or Locality, Measurement Year, PM2.5, PM10, NO2, % time coverage — one row per (city, year).

Eurostat — GDP, estat_nama_10_pc.tsv

freq, unit, na_item, geo, then one year per column (1975–2023): *wide* format.

Eurostat — Population, estat_demo_pjan.tsv

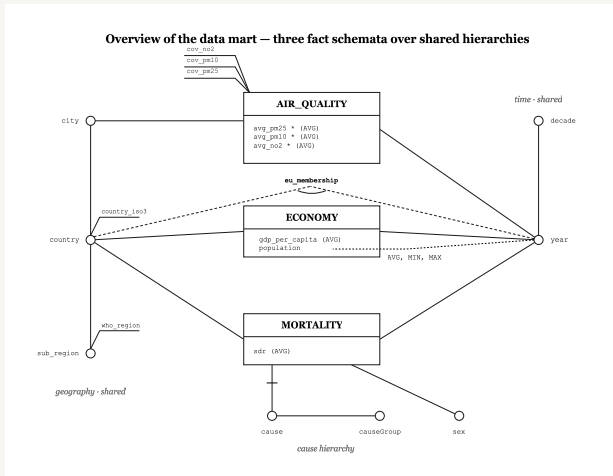
freq, unit, age, sex, geo, then one year per column: a separate file from GDP, with different columns — they are only joined at the reconciled stage.

WHO — HFAMDB Mortality (CSV, 4 files)

COUNTRY, COUNTRY_GRP, AGE_GRP_LIST, SEX, SUBNATIONAL_MDB, YEAR, VALUE: one row per (country, sex, year), one file per cause of death.

THE CONCEPTUAL MODEL (DFM)

- **Air Quality:** granularity City / Year. Measures include PM2.5, PM10, and NO2 levels, along with their respective temporal coverage.
- **Economy:** granularity Country / Year. Measures include GDP per capita and population.
- **Mortality:** granularity Country / Year / Gender / Cause. The measure is the Standardized Death Rate (SDR), it is aggregated with AVG, never with SUM. CAUSE is an optional dimension: populated only by the specific diseases.



THE CONSTELLATION MODEL

Why it is a *constellation* and not 3 separate schemas

TIME and GEOGRAPHY are **conformed dimensions**: shared by the 3 facts with the same identifiers and hierarchies. This is what enables **Drill-Across** queries – comparing avg_pm25, gdp_per_capita, and sdr by country and year, even though they come from different facts.

eu_membership is a **cross-dimensional** attribute: it depends on the combination (Country, Year), not on a single dimension – EU membership changes over time, as shown by two real transitions in the 2010–2020 window.

Country	Year	eu_membership
HRV	2012 → 2013	Non-EU → EU
GBR	2019 → 2020	EU → Non-EU

GRANULARITY AND BUS MATRIX

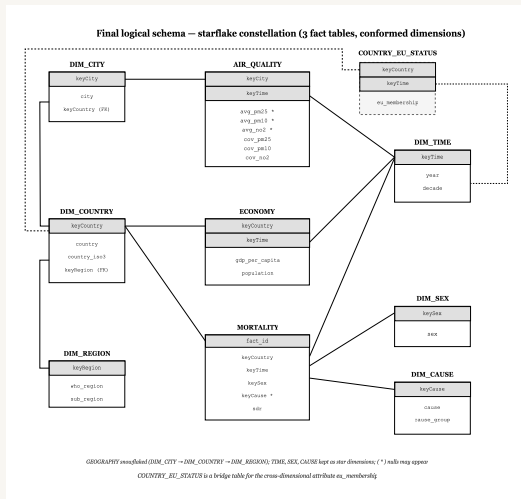
Fact	GEOGRAPHY	TIME	SEX	CAUSE
AIR_QUALITY	City	Year	-	-
ECONOMY	Country	Year	-	-
MORTALITY	Country	Year	Sex	Cause

■ Grain per fact and conformed dimensions

- Different grain per fact: AIR_QUALITY at **City** level (urban measurements), ECONOMY/MORTALITY at **Country** level (national data)
- Shared conformed dimensions: GEOGRAPHY and TIME (year → decade) enable cross-domain analysis

Implementation Details

- **Constellation structure** for the three fact tables.
- **Snowflake for Geography:**
The geographic hierarchy is normalized into three separate tables linked by foreign keys.
- **Star for the other dimensions:**
Time, Sex, and Cause are kept "flat" and denormalized in single tables.
- **Surrogate** keys assigned to each dimension table.



Why Snowflake for Geography

city → country → sub_region: avoids repeating the country name/ISO3 for hundreds of cities (redundancy), and handles the two different grain levels (city for air quality, country for economy/mortality) — a single flat DIM_GEOGRAPHY could not do this cleanly.

Why Star for Time, Sex, Cause

These are **small, static** tables (few rows, almost never updated): normalizing them like geography would add extra JOINS with no real benefit, since there is no significant redundancy to remove. Denormalization here is a pure win for reads.

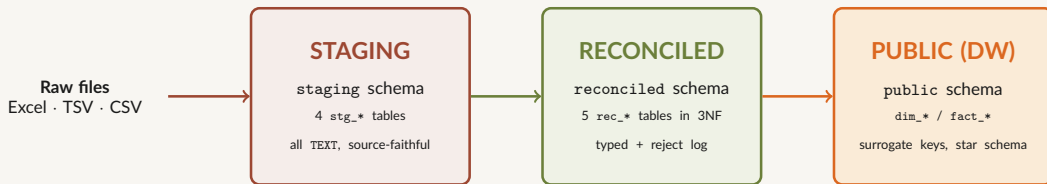
■ The problem

Only **3 specific diseases** belong to DIM_CAUSE. The total entry "*All causes*" is not a fourth cause: it is the overall total.

■ Solution: optional key

- "*All causes*" → key_cause = NULL in the MORTALITY fact
- Integrity guaranteed in PostgreSQL by **two partial unique indexes**:
uq_mortality_cause and uq_mortality_total
- They prevent double counting between totals and per-cause values

3-TIER ELT ARCHITECTURE



We adopt a **3-tier ELT**: raw data is loaded immediately, as-is, into the staging schema; cleaning and integration happen *afterwards*, with declarative SQL queries inside PostgreSQL, moving through the reconciled and public tiers.

TIER 1: STAGING

Python's role changes: only *ingestion*, never *cleaning*

Python reads Excel/TSV/CSV files and writes them *exactly as they are* into the staging schema: no parsing, no casting. Everything stays TEXT on purpose — a NUMERIC cast would immediately fail on Eurostat flags ("31200 p", ":"); casting becomes a decision postponed to the next step.

Domain	Staging table	Source
Air quality	stg_who_aap	1 WHO Excel file
GDP	stg_eurostat_gdp	1 Eurostat TSV file
Population	stg_eurostat_pop	1 Eurostat TSV file
Mortality	stg_hfamdb_mortality	4 HFAMDB CSV files (UNION)

■ From *staging* to *reconciled*: cleaning and integration

- **Cleaning:** non-interpretable values are discarded; *wide* format (one year per column) → *long* format (one row per year)
- **Integration:** GDP and Population (2 separate Eurostat files) are merged into a single row per country-year, without losing data when one source does not cover a given country/year
- Tables in **3NF: natural** keys (e.g. `country_iso3`), constraints (PK, FK, CHECK), and specific types instead of the TEXT used in staging

reconciled.* table	Grain (1 row per ...)
rec_country	Country
rec_city	(Country, City)
rec_air_quality	(Country, City, Year)
rec_economy	(Country, Year)
rec_mortality	(Country, Year, Sex, Cause)

Every discarded row leaves a trace

Every rejection creates an explicit row in `reconciled.rec_reject_log`: table, key, reason, and timestamp — so we can reconstruct what was excluded and why, without re-running the pipeline.

Reason for rejection	Example
Eurostat aggregate, not a country	EU27_2020, EA20
ISO3 code outside geographic Europe	AFG, USA

`v_reject_summary`: the aggregated log

A view that groups the log by table and reason, counting occurrences: one query answers "how many rows were excluded, and why".

■ From reconciled to public: surrogate key lookup

The `reconciled.*` tables feed the dimensions and facts, replacing natural keys (ISO3, city, year) with surrogate keys. At the end of the ELT process, the `public` schema contains the fully populated fact constellation, ready for analytical queries.

From reconciled.*	To public.*
<code>rec_country</code>	<code>dim_region, dim_country</code>
<code>rec_city</code>	<code>dim_city</code>
<code>rec_time</code>	<code>dim_time</code>
<code>rec_air_quality</code>	<code>air_quality (fact)</code>
<code>rec_economy</code>	<code>economy (fact), country_eu_status</code>
<code>rec_mortality</code>	<code>mortality (fact)</code>

For mortality, the *"All causes"* row does not match any specific cause: `key_cause` automatically stays NULL.

■ Inconsistent country codes

WHO/HFAMDB use **ISO3** (3 letters), Eurostat uses **geo** with **2 letters** (2 exceptions: EL = Greece, UK = United Kingdom): a dedicated mapping table is needed to bring them all back to one common key.

- `sub_region` (Northern/Southern/Eastern/Western Europe): does not exist in the original datasets, defined by hand to enable geographic roll-up
- `cause_group` (e.g. *cardiopulmonary*, *cancer*): the 3 specific causes of death were grouped from scratch for a more natural drill-down
- Cleaning numbers: Eurostat flags ("31200 p" = provisional, ":" = missing) are cleaned before casting to numeric

ROLL-UP

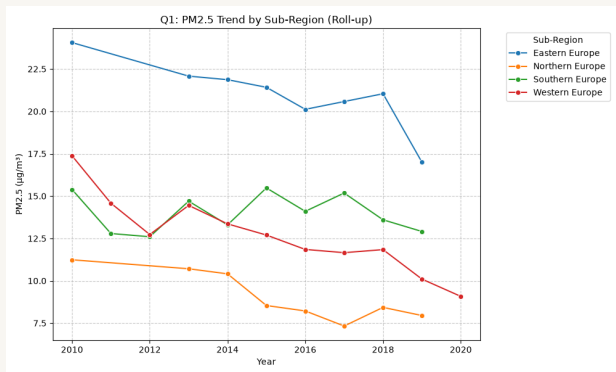
```
SELECT r.sub_region,
       t.year,
       COUNT(DISTINCT ci.key_city)      AS n_cities,
       ROUND(AVG(aq.avg_pm25), 2)      AS pm25,
       ROUND(AVG(aq.avg_pm10), 2)      AS pm10,
       ROUND(AVG(aq.avg_no2), 2)      AS no2
FROM   air_quality aq
JOIN   dim_city    ci ON ci.key_city    = aq.key_city
JOIN   dim_country co ON co.key_country = ci.key_country
JOIN   dim_region  r  ON r.key_region   = co.key_region
JOIN   dim_time    t  ON t.key_time     = aq.key_time
GROUP BY r.sub_region, t.year
ORDER BY r.sub_region, t.year;
```

Roll-up query that aggregates air quality data. Air quality is stored at city grain. We climb the geographic hierarchy city → country → sub_region and aggregate to the region level, by year.

ROLL-UP INSIGHT: PM2.5 TREND

Roll-up

Trend of PM2.5 levels over time across different sub-regions in Europe.



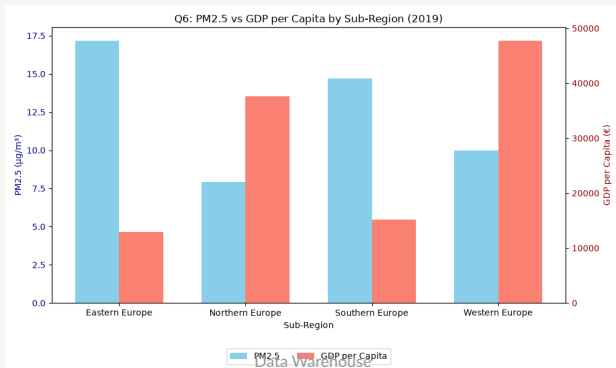
DRILL-ACROSS

```
WITH aq_country AS (  
    SELECT ci.key_country, aq.key_time,  
           AVG(aq.avg_pm25) AS pm25, AVG(aq.avg_no2) AS no2, COUNT(*) AS n_cities  
    FROM   air_quality aq  
    JOIN   dim_city ci ON ci.key_city = aq.key_city  
    GROUP BY ci.key_country, aq.key_time  
) ,  
resp AS (  
    SELECT m.key_country, m.key_time, m.sdr  
    FROM   mortality m  
    JOIN   dim_cause ca ON ca.key_cause = m.key_cause  
    JOIN   dim_sex s ON s.key_sex = m.key_sex  
    WHERE  ca.cause = 'Respiratory diseases' AND s.sex = 'ALL'  
)  
SELECT r.sub_region, t.year,  
       COUNT(DISTINCT a.key_country) AS n_countries,  
       SUM(a.n_cities) AS n_cities,  
       ROUND(AVG(a.pm25)::numeric, 2) AS pm25,  
       ROUND(AVG(re.sdr)::numeric, 2) AS respiratory_sdr,  
       ROUND(AVG(e.gdp_per_capita)::numeric, 0) AS gdp_per_capita  
FROM   aq_country a  
JOIN   dim_country co ON co.key_country = a.key_country  
JOIN   dim_region r ON r.key_region = co.key_region  
JOIN   dim_time t ON t.key_time = a.key_time  
LEFT JOIN resp re ON re.key_country = a.key_country AND re.key_time = a.key_time  
LEFT JOIN economy e ON e.key_country = a.key_country AND e.key_time = a.key_time  
GROUP BY r.sub_region, t.year  
HAVING COUNT(DISTINCT a.key_country) >= 3  
ORDER BY r.sub_region, t.year;
```

POLLUTION AND WEALTH: THE CONSTELLATION AT WORK

Drill-Across

Joining AIR_QUALITY and ECONOMY on the conformed dimensions GEOGRAPHY and TIME: in 2019 Eastern Europe has the highest PM2.5 ($17.2 \mu\text{g}/\text{m}^3$) and the lowest GDP per capita (€12,970) – Western Europe shows the exact opposite ($10.0 \mu\text{g}/\text{m}^3$, €47,767).



■ In summary

- An effective **ROLAP Constellation** architecture for connecting heterogeneous domains (air, economy, health)
- DFM design → Starflake logical schema → 3-tier ELT pipeline (staging → reconciled → public), reproducible and auditable
- Advanced OLAP analyses reveal the link between pollution, GDP, and mortality

Moving to the live demo